

หลักฐานการลงมือทำ

สกรีนช็อตประกอบ Learning Log — Linux CLI 5-Day Challenge

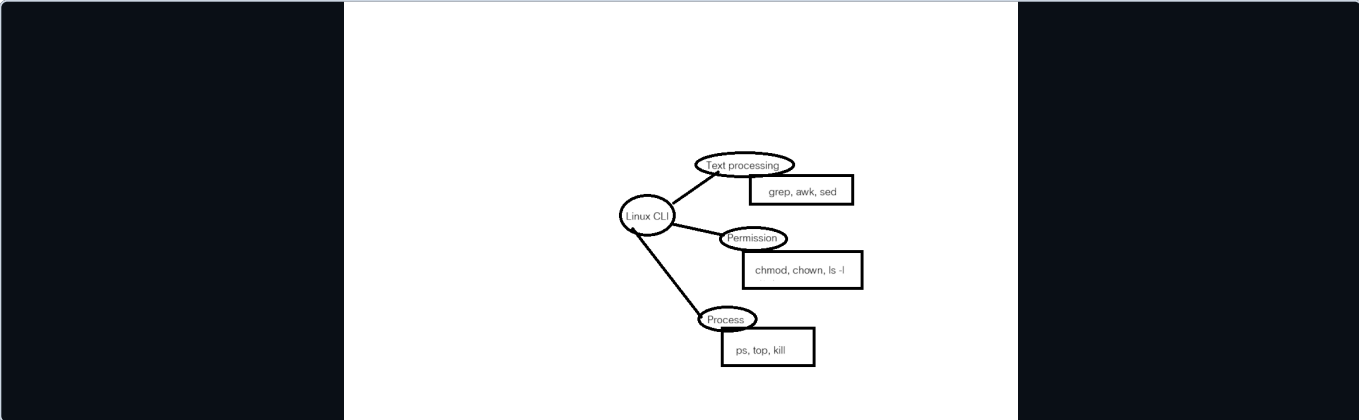
Day 1 — Setup

Push โค้ดขึ้น GitHub repository สำเร็จครั้งแรก (branch main -> main)

```
Username for 'https://github.com': ftostbit-dot
Password for 'https://ftostbit-dot@github.com':
remote: Invalid username or token. Password authentication is not supported for Git open
fatal: Authentication failed for 'https://github.com/ftostbit-dot/linux-cli-challenge.git'
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ git p
Username for 'https://github.com': ftostbit-dot
Password for 'https://ftostbit-dot@github.com':
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 20 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (16/16), 6.59 KiB | 129.00 KiB/s, done.
Total 16 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ftostbit-dot/linux-cli-challenge.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ |
```

Day 2 — Concept Map และ Tutorial Walkthrough

Concept Map แบ่ง Linux CLI เป็น 3 สิ่งหลัก (Text Processing, Permission, Process)



Push concept map ขึ้น GitHub สำเร็จ

```
Ftost@Ftost: /mnt/c/Users/LE x + v
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\LEGION>wsl
Ftost@Ftost:/mnt/c/Users/LEGION$ cd /mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge
git add -A
git commit -m "Add day 2 concept map"
git push
[main 727c5ac] Add day 2 concept map
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 day2/2daypic.png
Username for 'https://github.com': ftostbit-dot
Password for 'https://ftostbit-dot@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 19.51 KiB | 1.50 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ftostbit-dot/linux-cli-challenge.git
e422baa..727c5ac main -> main
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ |
```

ทดสอบ pipe: grep "Failed" | wc -l ได้ผลลัพธ์ 26 (พร้อม error wc -l ที่เออเรอข้างทาง)

```
Ftost@Ftost: /mnt/c/Users/LE x + v
ssh2
Sep 1 14:22:18 web01 sshd[1130]: Failed password for invalid user git from 91.240.118.77 port 41200 ssh2
Sep 1 14:22:22 web01 sshd[1131]: Failed password for invalid user git from 91.240.118.77 port 41210 ssh2
Sep 1 14:22:26 web01 sshd[1132]: Failed password for invalid user ubuntu from 91.240.118.77 port 41225 ssh2
Sep 1 22:41:09 web01 sshd[1170]: Failed password for admin from 45.155.204.19 port 23301 ssh2
Sep 1 22:41:13 web01 sshd[1171]: Failed password for admin from 45.155.204.19 port 23320 ssh2
Sep 1 22:41:17 web01 sshd[1172]: Failed password for root from 45.155.204.19 port 23345 ssh2
Sep 1 22:41:21 web01 sshd[1173]: Failed password for root from 45.155.204.19 port 23360 ssh2
Sep 1 22:41:25 web01 sshd[1174]: Failed password for root from 45.155.204.19 port 23388 ssh2
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ grep "Failed" logs/auth.log | wc -l
error: unexpected argument '-l' found
tip: to pass '-l' as a value, use '-- -l'
Usage: wc [OPTION]... [FILE]...
For more information, try '--help'.
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ grep "Failed" logs/auth.log | wc -l
26
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ |
```

Day 3 — Log Analysis

หา IP ที่พยายาม login ล้มเหลวมากที่สุด (grep -oE | sort | uniq -c) — 185.220.101.5 ติดอันดับ 1

```
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ grep "Failed" logs/auth.log | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | sort | uniq -c
10 185.220.101.5
8 45.155.204.19
8 91.240.118.77
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ |
```

Error 2: awk '{print \$11}' ได้ค่า "git"/"ubuntu" หมายความว่า IP เพราะค่าล้นไป

```
8 91.240.118.77
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ grep "Failed" logs/auth.log | awk '{print $9}' | sort | uniq -c
10 admin
3 invalid
2 oracle
2 postgres
7 root
2 test
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ |
```

แก้ปัญหาด้วย grep -v "invalid" ก่อน awk — ผลลัพธ์สะอาดขึ้น

```
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ grep "Failed" logs/auth.log | grep -v "invalid" | awk '{print $9}' | sort | uniq -c
  10 admin
   2 oracle
   2 postgres
   7 root
   2 test
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ |
```

แยกบรรทัด login สำเร็จเป็นไฟล์ successful_logins.txt

```
PRACTICE_TASKS.md README.md WSL_SETUP.md Day2 Day3 Day4 Day5 Logs successful_logins.txt
Sep  1 02:14:03 web01 sshd[1021]: Accepted password for front from 203.113.45.10 port 51322 ssh2
Sep  1 02:20:11 web01 sshd[1030]: Accepted password for front from 203.113.45.10 port 51890 ssh2
Sep  1 07:44:12 web01 sshd[1090]: Accepted publickey for deploy from 203.113.45.10 port 52210 ssh2
Sep  1 11:30:02 web01 sshd[1110]: Accepted password for front from 203.113.45.10 port 55012 ssh2
Sep  1 18:05:44 web01 sshd[1150]: Accepted password for front from 203.113.45.10 port 58800 ssh2
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ |
```

Day 3 — Permission Practice

Error 3: chmod 600 ไม่มีผลกับไฟล์ใน /mnt/c/ (Windows filesystem)

```
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ chmod 600 secret_config.txt
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ ls -l secret_config.txt
-rwxrwxrwx 1 Ftost Ftost 0 Sep  8 17:19 secret_config.txt
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ |
```

ย้ายไปทดสอบใน Linux home directory — permission default เป็น rw-r--r--

```
Ftost@Ftost:/mnt/c/Users/LEGION/Downloads/linux-cli-challenge/linux-cli-challenge$ cd ~
touch secret_config.txt
ls -l secret_config.txt
-rw-r--r-- 1 Ftost Ftost 0 Sep  8 17:22 secret_config.txt
Ftost@Ftost:~$ |
```

chmod 600 สำเร็จ — permission เปลี่ยนเป็น rw----- ตามที่ตั้งใจ

```
Ftost@Ftost:~$ chmod 600 secret_config.txt
ls -l secret_config.txt
-rw----- 1 Ftost Ftost 0 Sep  8 17:22 secret_config.txt
Ftost@Ftost:~$ |
```

Day 3 — Process Practice

รัน sleep 300 & เสร็จแล้ว แล้วหา process ด้วย ps aux | grep sleep — เจอ PID 977

```
Ftost@Ftost:~$ sleep 300 &
[1] 977
Ftost@Ftost:~$ ps aux | grep sleep
Ftost   977  0.0  0.0 16112 7832 pts/0    S   17:24   0:00 sleep 300
Ftost   988  0.0  0.0  4128 2400 pts/0    S+  17:24   0:00 grep --color=auto sleep
Ftost@Ftost:~$ |
```

kill 977 สำเร็จ — process ถูก Terminated และหายไปจากรายการเมื่อเช็คซ้ำ

```
Ftost@Ftost:~$ kill 977
[1]+  Terminated                  sleep 300
Ftost@Ftost:~$ ps aux | grep sleep
Ftost   1009  0.0  0.0  4128 2436 pts/0    S+  17:25   0:00 grep --color=auto sleep
Ftost@Ftost:~$ |
```